ACTIVITY-04

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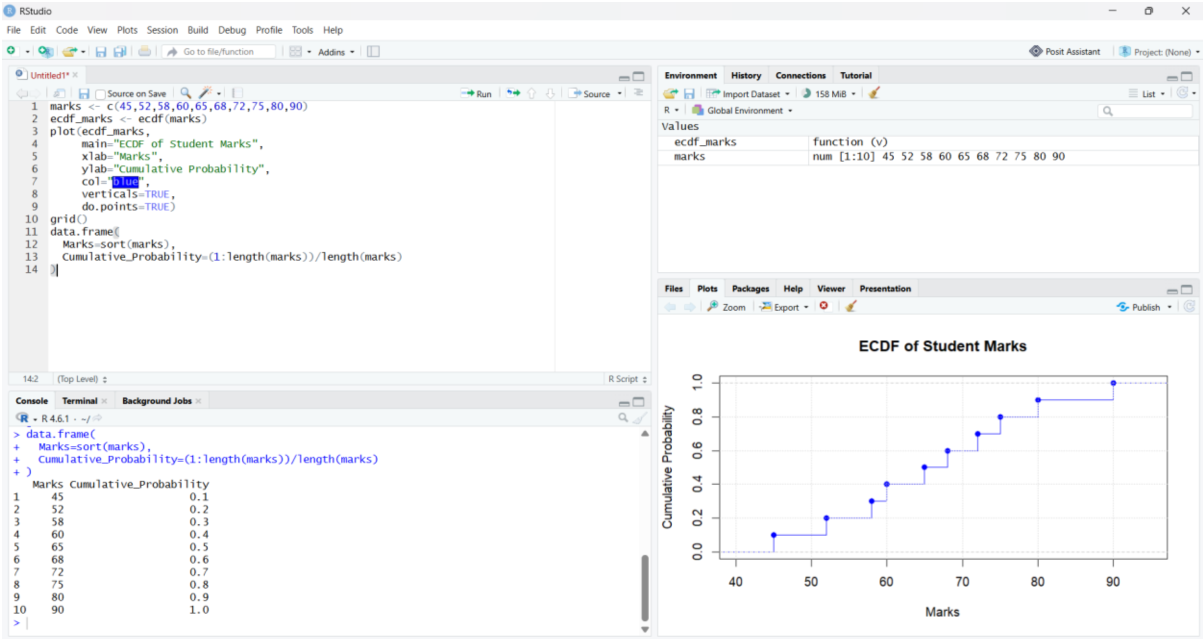
Q1. **Empirical Cumulative Distribution Function (ECDF)**

Program:

```
marks <- c(45,52,58,60,65,68,72,75,80,90)

ecdf_marks <- ecdf(marks)

plot(ecdf_marks,

    main="ECDF of Student Marks",

    xlab="Marks",

    ylab="Cumulative Probability",

    col="blue",

    verticals=TRUE,

    do.points=TRUE)

grid()

data.frame(

  Marks=sort(marks),

  Cumulative_Probability=(1:length(marks))/length(marks)

)
```

OUTPUT:

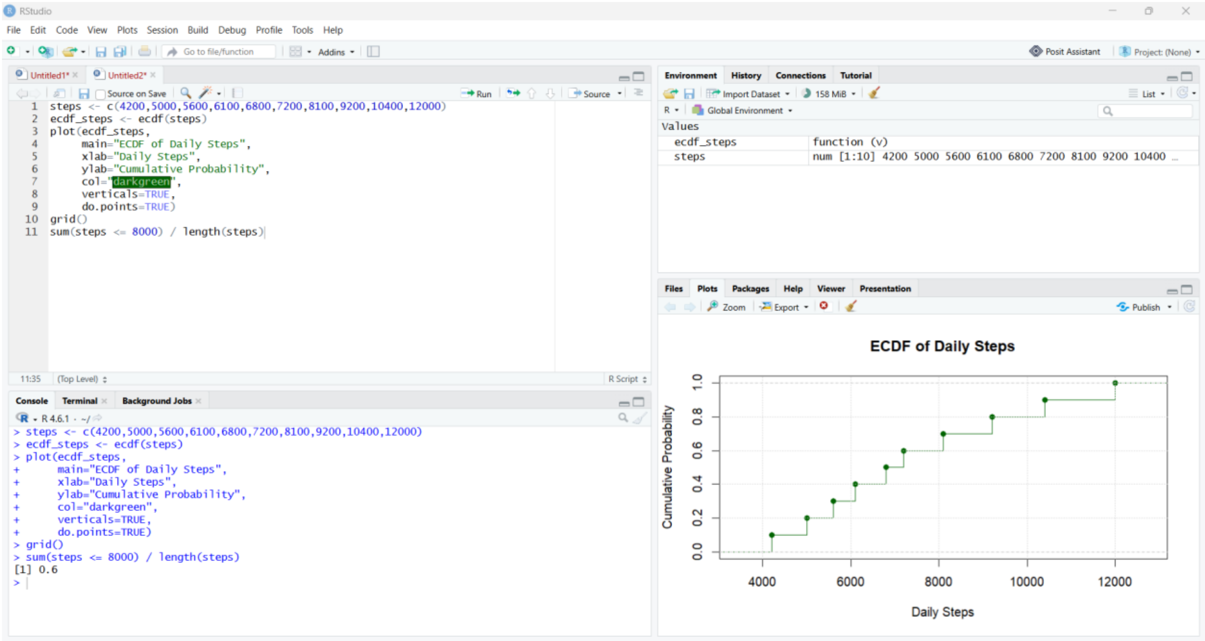


Q2. ECDF (Daily Steps)

PROGRAM:

```
steps <- c(4200,5000,5600,6100,6800,7200,8100,9200,10400,12000)

ecdf_steps <- ecdf(steps)

plot(ecdf_steps,

    main="ECDF of Daily Steps",

    xlab="Daily Steps",

    ylab="Cumulative Probability",

    col="darkgreen",

    verticals=TRUE,

    do.points=TRUE)

grid()

sum(steps <= 8000) / length(steps)
```

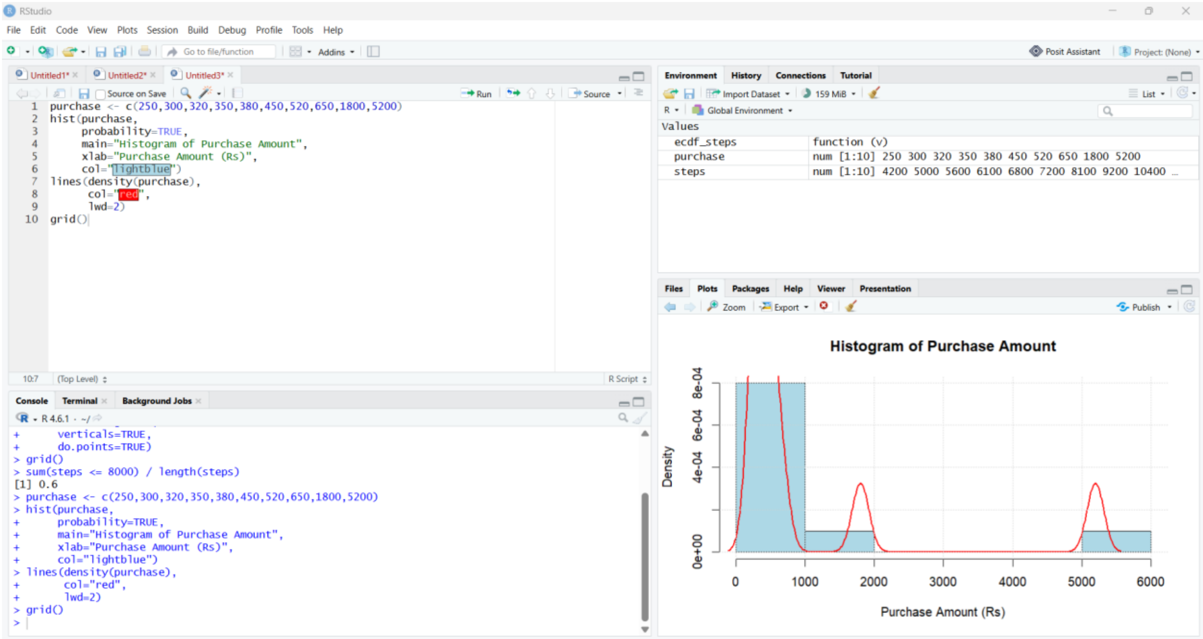
PROGRAM:



Q3. Histogram and Density Plot (Purchase Amount)

```
purchase <- c(250,300,320,350,380,450,520,650,1800,5200)

hist(purchase,

     probability=TRUE,

     main="Histogram of Purchase Amount",

     xlab="Purchase Amount (Rs)",

     col="lightblue")

lines(density(purchase),

      col="red",

      lwd=2)

grid()
```

OUTPUT:

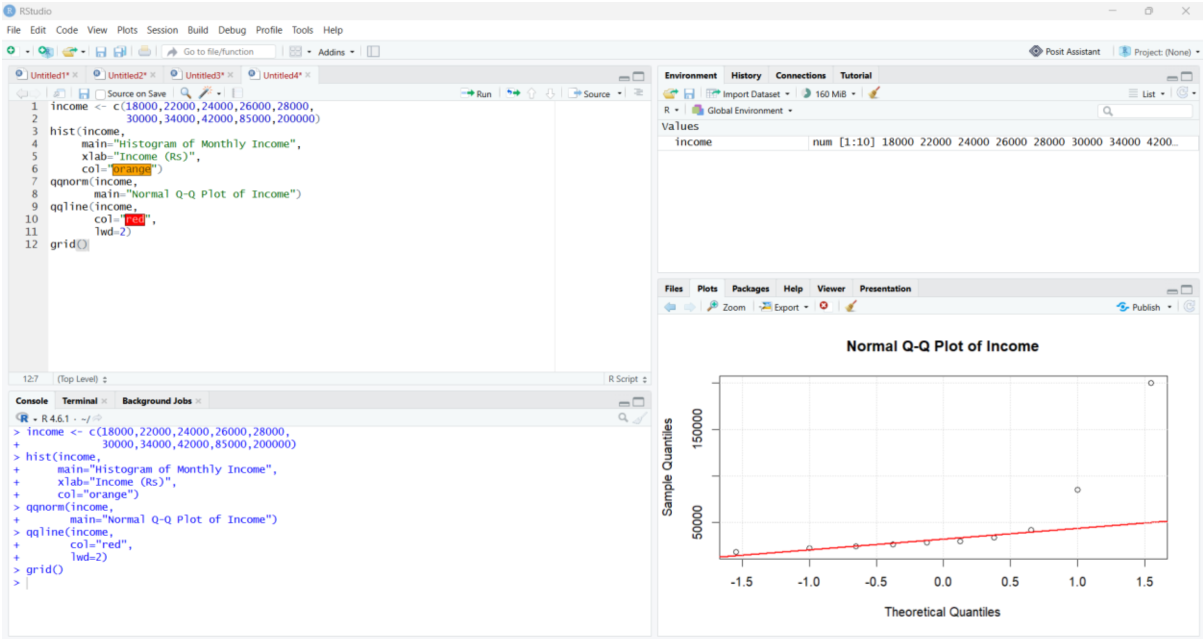


Q4. **Normal Q-Q Plot (Monthly Income)**

Program:

```
income <- c(18000,22000,24000,26000,28000,

        30000,34000,42000,85000,200000)

hist(income,

   main="Histogram of Monthly Income",

   xlab="Income (Rs)",

   col="orange")

qqnorm(income,

     main="Normal Q-Q Plot of Income")

qqline(income,

     col="red",

     lwd=2)

grid()
```

PROGRAM:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1* Untitled2* Untitled3* Untitled4*
income <- c(18000,22000,24000,26000,28000,
30000,34000,42000,85000,200000)
hist(income,
main="Histogram of Monthly Income",
xlab="Income (Rs)",
col="orange")
qqnorm(income,
main="Normal Q-Q Plot of Income")
qqline(income,
col="red",
lwd=2)
grid()
Console Terminal Background Jobs
R 4.6.1
Environment History Connections Tutorial
Import Dataset
Global Environment
Values
income num [1:10] 18000 22000 24000 26000 28000 30000 34000 4200…
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Normal Q-Q Plot of Income
Sample Quantiles
50000
150000
-1.5 -1.0 -0.5 0.0 0.5 1.0 1.5
Theoretical Quantiles